GROUP 27

MOVIES VS POLITICS

OPEN PROJECT PRESENTATION

THE PROBLEM AND DATA

PROBLEM STATMENT: **We plan to analyse movies over the past decades using their genres like action, thriller, romance, comedy, drama, horror, etc., against the political climate. We will be measuring the political climate and uncertainty with crime rates, protest numbers, and the margin of votes for the ruling party (to measure political dissatisfaction). We will also be focusing on standout politically tense times, such as wars.**



THE RANDOM VARIABLES

Fraction values of number of movies per genre for every genre ever year

$$E_t = 1 - (V_t / 100)$$

$$P_t = (\text{Protests}_t - \text{Protests}_{min}) / (\text{Protests}_{max} - \text{Protests}_{min})$$

W_t = 1 if war/military operation occurred in that year; 0 otherwise

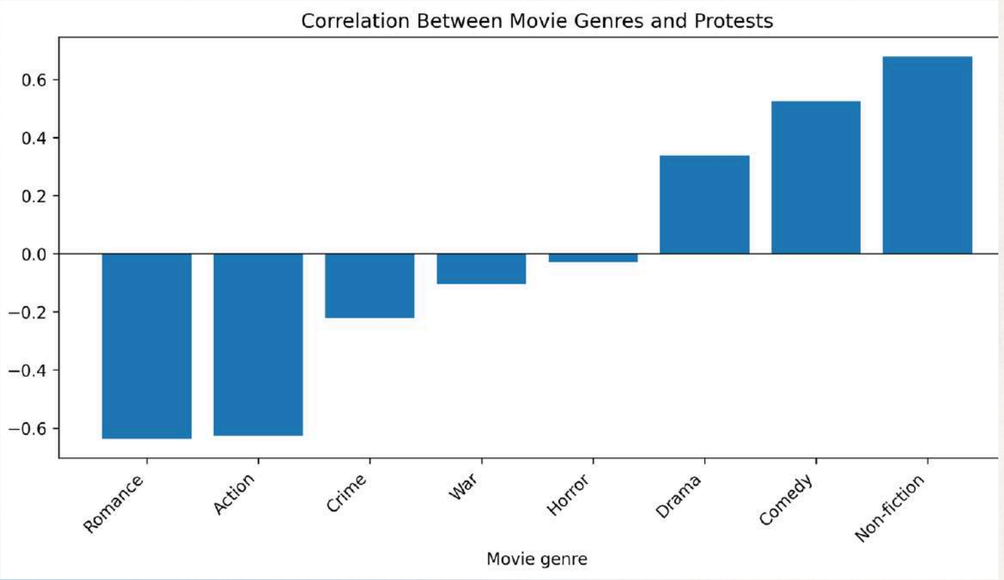
$$C_t = (\text{Crimes}_t - \text{Crimes}_{min}) / (\text{Crimes}_{max} - \text{Crimes}_{min})$$

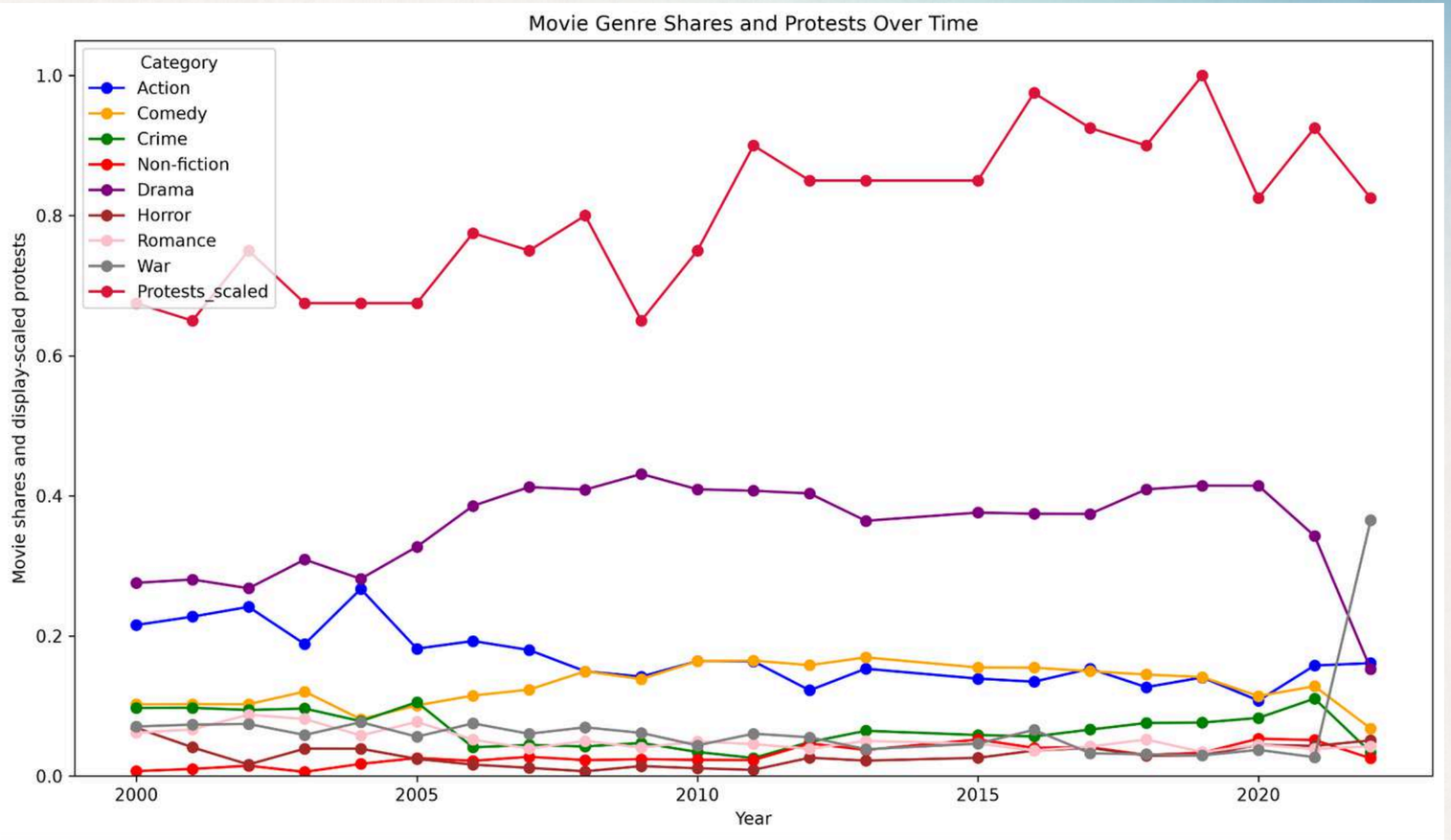
$$PUC_t = 100 \times (0.15E_t + 0.40C_t + 0.35P_t + 0.10W_t)$$

POLITICAL UNCERTAINTY COEFFICIENT

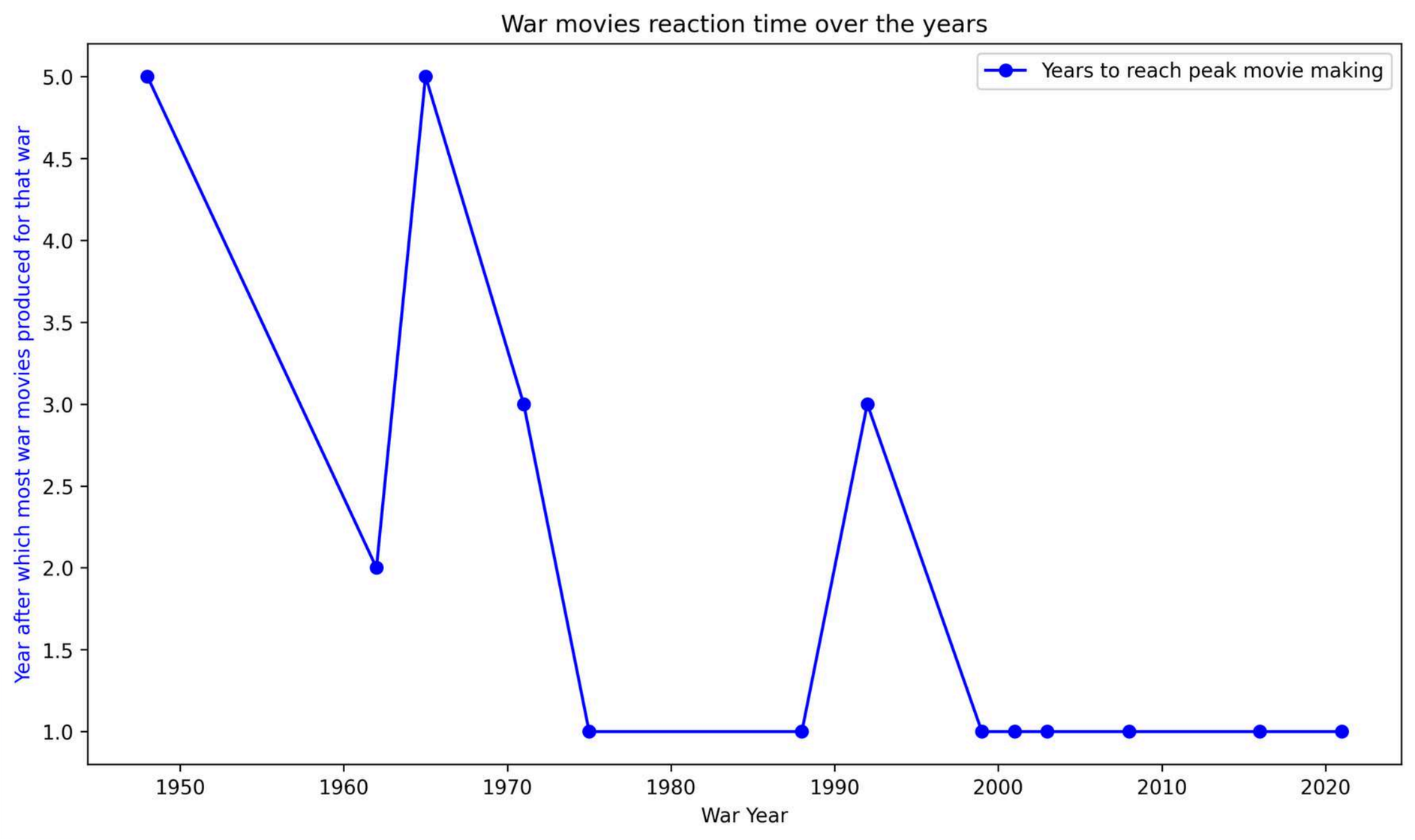
PROTEST VS NON-FICTION

corelation coefficient = 0.679



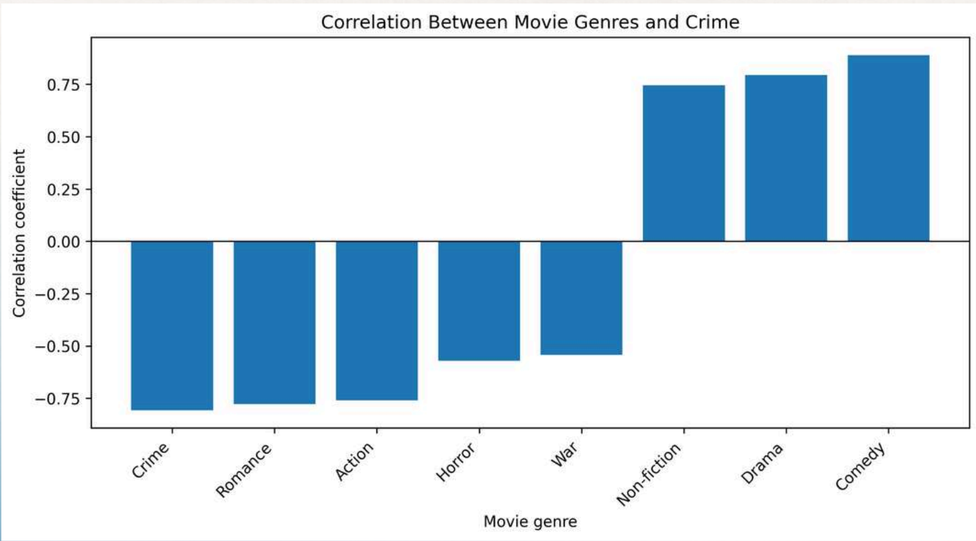


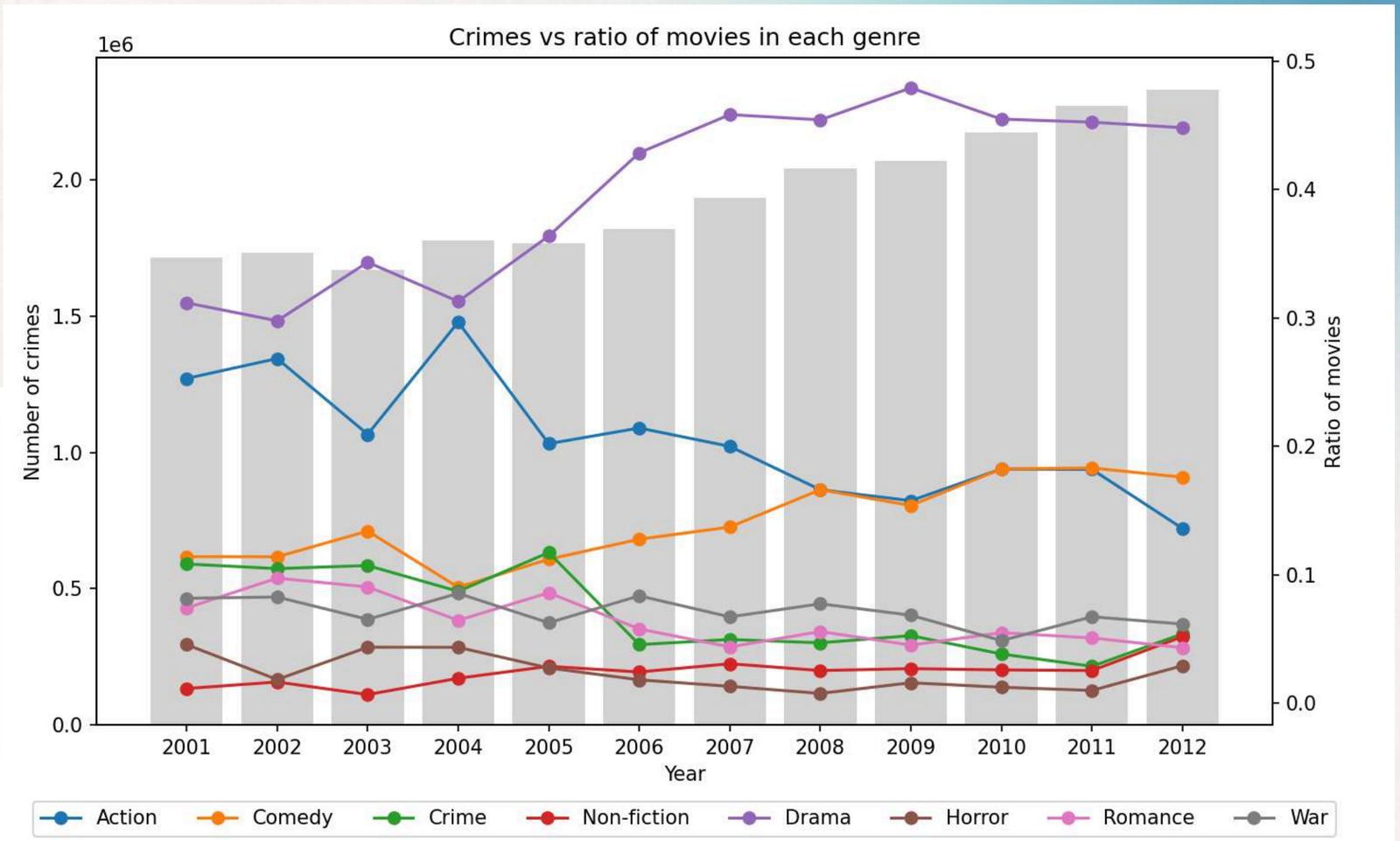
WAR REACTION TIME



CRIME VS COMEDY MOVIES

corelation coefficient = 0.89
Interestingly the corelation coefficient between crime in India and crime-related movies is negative.
corelation coefficient = -0.807

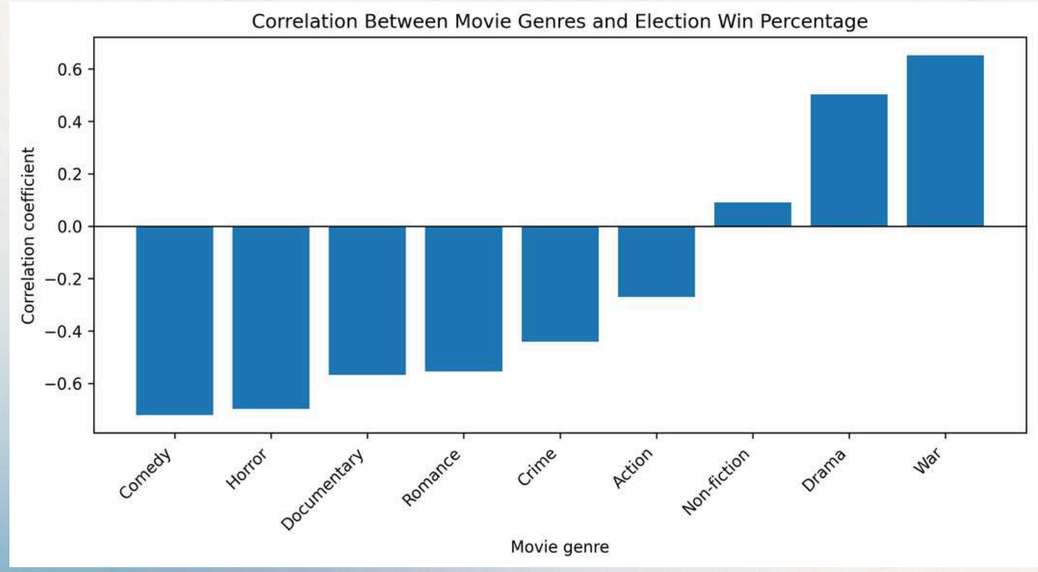


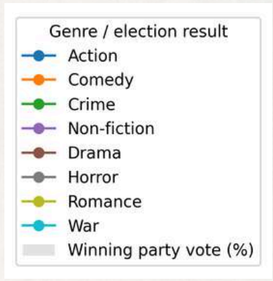


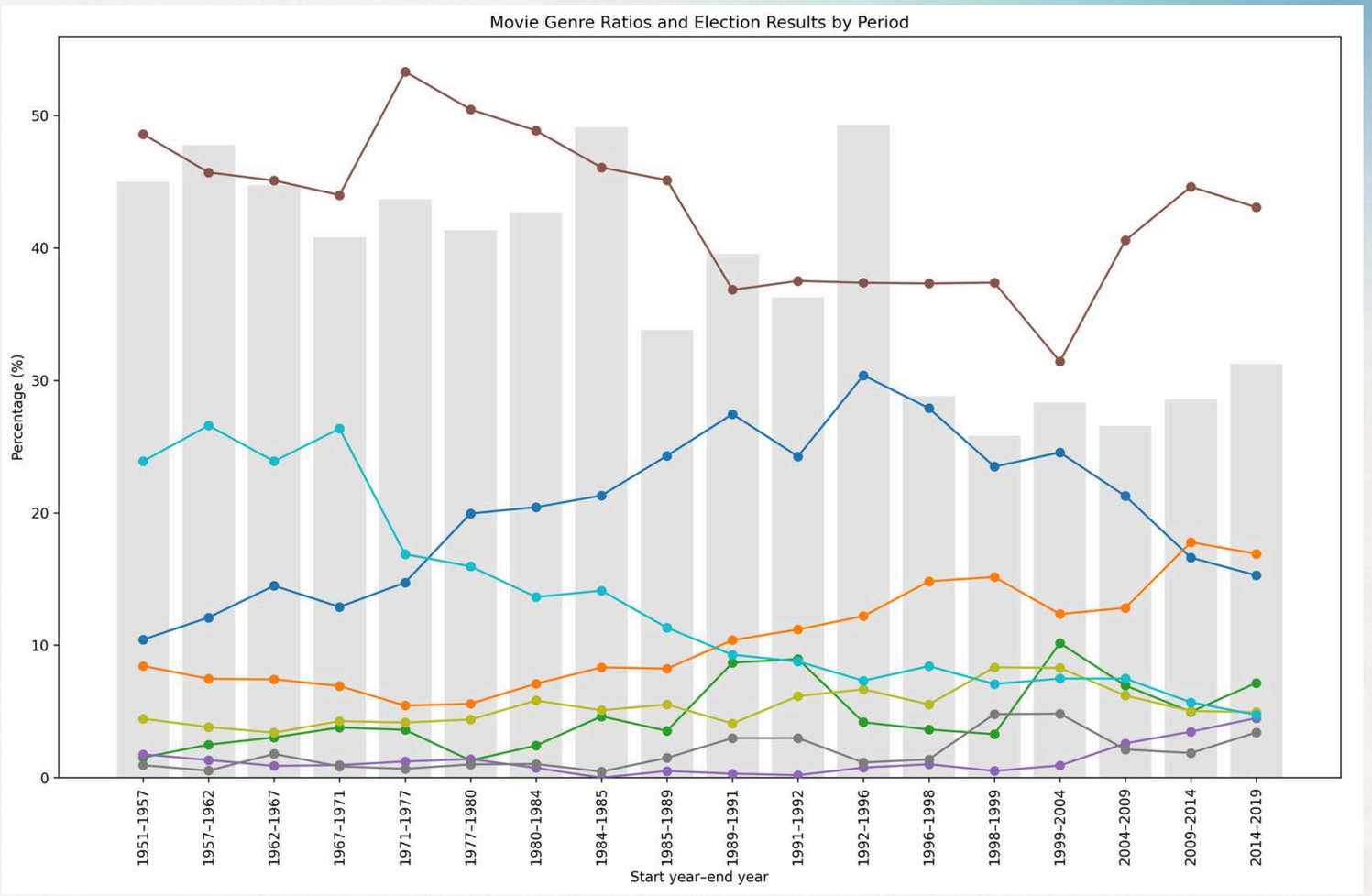
ELECTION VS COMEDY MOVIES

corelation coefficient = -0.721

This observation maybe attributed to the fact that audiences often prefer lighter movies in heavier social situations. It can also be rooted in the fact that some comedy movies centre around mocking political parties







REGRESSION ANALYSIS:

CRIMES

Genre	slope_0	intercept_0	r_squared_0	slope_1	intercept_1	r_squared_1
Drama	0.024	-0.058	0.63	0.024	-0.049	0.538
Action	-0.016	0.507	0.577	-0.019	0.56	0.658
Comedy	0.012	-0.092	0.792	0.014	-0.121	0.807
Crime	-0.011	0.289	0.652	-0.01	0.267	0.455
Romance	-0.006	0.188	0.605	-0.006	0.186	0.446
Non-fiction	0.004	-0.046	0.557	0.004	-0.047	0.484
Horror	-0.003	0.09	0.325	-0.003	0.074	0.181
War	-0.003	0.122	0.295	-0.003	0.132	0.336

01

Action has negative slopes for crimes (-0.016), elections (-0.098) and protests (-0.065). The Audience seems to move away from action when political uncertainty rises. Drama is the opposite with positive slopes in all three, so Drama may be the comfort genre during political strain.

ELECTIONS

Genre	slope_0	intercept_0	r_squared_0	slope_1	intercept_1	r_squared_1
War	0.295	-0.092	0.425	0.284	-0.085	0.43
Drama	0.177	0.295	0.253	0.156	0.31	0.199
Comedy	-0.171	0.234	0.519	-0.161	0.228	0.452
Action	-0.098	0.275	0.073	-0.042	0.233	0.015
Crime	-0.071	0.1	0.194	-0.095	0.119	0.367
Horror	-0.059	0.064	0.488	-0.059	0.064	0.482
Romance	-0.048	0.09	0.308	-0.047	0.089	0.294
Documentary	-0.03	0.027	0.323	-0.036	0.032	0.447
Non-fiction	0.003	0.006	0.008	-0.0	0.008	0.0

02

Protest numbers are the weakest predictors with average R^2 value 0.21 (in contrast to 0.55 of CRIMES and 0.29 of ELECTIONS. We can conclude protest numbers explain genre trends less than crimes and elections do.

PROTESTS

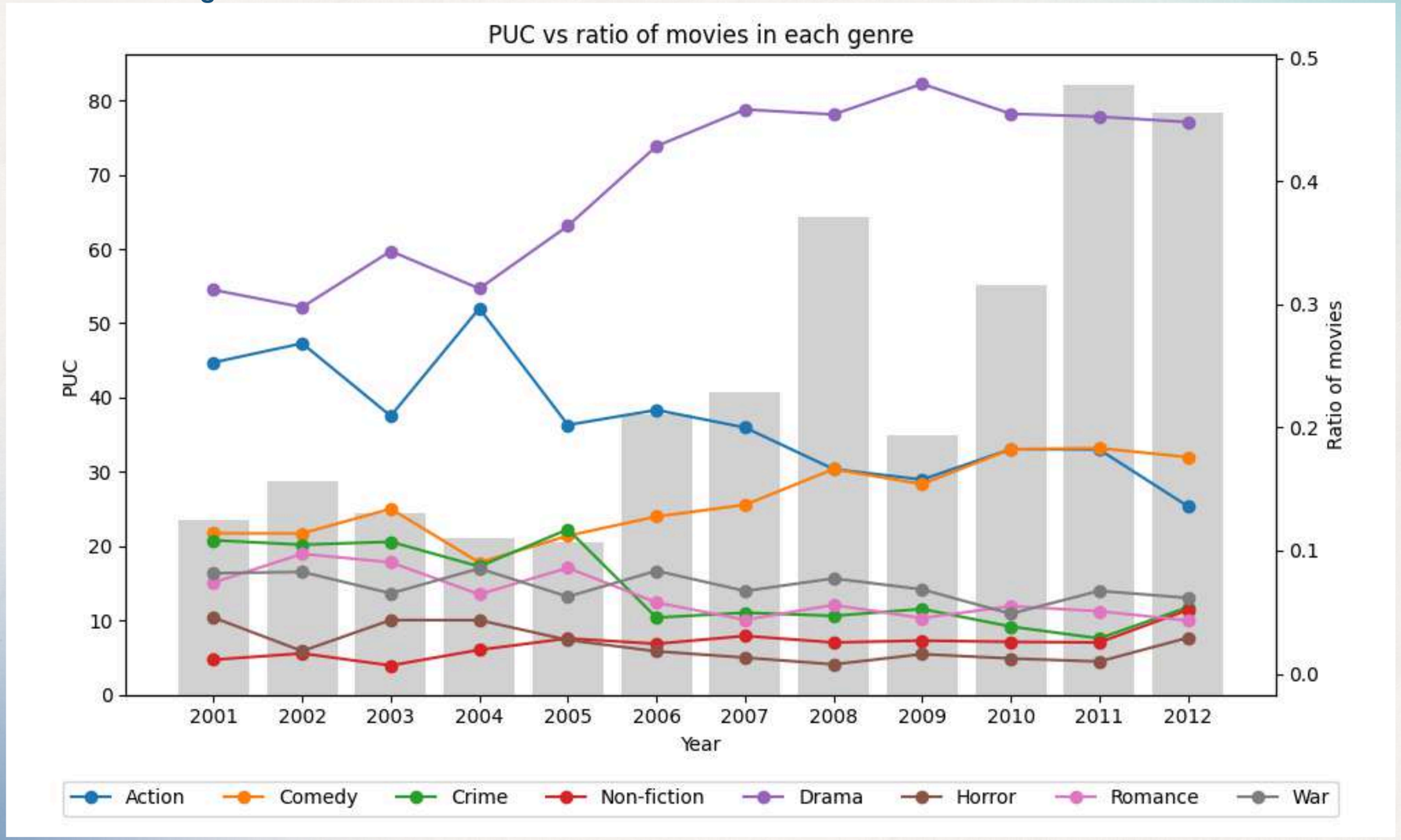
Genre	slope_0	intercept_0	r_squared_0	slope_1	intercept_1	r_squared_1
Action	-0.065	0.233	0.393	-0.076	0.241	0.579
Drama	0.06	0.353	0.114	0.032	0.373	0.034
Comedy	0.038	0.117	0.276	0.017	0.132	0.054
Non-fiction	0.025	0.015	0.461	0.024	0.015	0.503
Romance	-0.024	0.075	0.406	-0.023	0.074	0.372
War	-0.018	0.089	0.011	0.016	0.065	0.008
Crime	-0.015	0.085	0.049	-0.005	0.078	0.006
Horror	-0.001	0.034	0.001	0.016	0.022	0.222

03

Crime films have a negative slope against actual crime rates. As real-world crimes rise, viewers move away from crime-themed content. We conclude Audiences don't seek out genres that mirror their reality.

MOVIES VS PUC :)

The graph shows direct corelation between comedy (cc= 0.872) and drama movies (cc = 0.794) with the puc and a negative corelation with crime (cc = -0.615) and action movies (-0.027). This shows that the population prefers lighter movies that are almost fantasy-like when real life gets uncertain.



GROUP 27

THANK YOU VERY MUCH

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